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**SACUBITRIL INTERMEDIATE AND
PREPARATION METHOD THEREOF****CROSS-REFERENCE TO RELATED
APPLICATIONS**

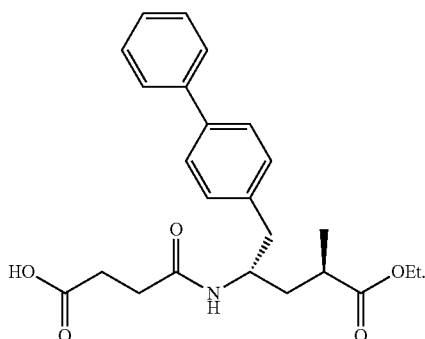
This is a divisional application of the U.S. patent application Ser. No. 16/112,761, filed Aug. 26, 2018, which is a continuation-in-part application of the International Patent Application No. PCT/CN2017/075097, filed Feb. 28, 2017, which claims priority to Chinese Patent Application No. 201610111396.6, filed Feb. 29, 2016, all of which are incorporated herein by reference in their entirety.

FIELD OF THE INVENTION

The present invention relates to the medical and chemical industry field, more specifically to a sacubitril intermediate and a preparation method thereof.

BACKGROUND OF THE INVENTION

Sacubitril, the chemical name is 4-(((2S,4R)-1-([1,1'-biphenyl]-4-yl)-5-ethoxy-4-methyl-5-oxopentan-2-yl)amino)-4-oxobutanoic acid, the structure is as following:



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U.S. FDA approved the anti-heart failure (anti-HF) drug Entresto (known as LCZ696) on July, 2015, which is a complex of the angiotensin II inhibitor valsartan and the
5 enkephalinase inhibitor sacubitril. Valsartan can improve vasodilation and stimulate the body to excrete sodium and water, and sacubitril can block the mechanism of action of two peptides that threaten to lower blood pressure, thus
10 LCZ696 is known as a dual inhibitor of angiotensin II receptor and enkephalinase, its clinical manifestation is a unique mode of action, which is superior to the standard
15 drugs with respect to antihypertensive effect and the efficacy of reducing heart failure.

Some methods of preparing sacubitril are disclosed in the
20 prior art, such as:

Patent application WO2008031567A1 disclosed a method
of preparing biphenyl butyric acid substituted on 4 position
25 amino or derivative thereof by using biphenyl propanol substituted on 2 position amino as starting materials, which was oxidized to aldehyde, and the aldehyde was suffered
30 from Witting reaction to give biphenyl butenoic acid substituted on 4 position amino or derivative thereof, and the biphenyl butenoic acid substituted on 4 position amino or
35 derivative thereof was suffered from a reduction reaction under H₂ in the presence of transition metal of group 7, 8 and 9 in the chemical periodic table as catalyst and a chiral
ligand, and then chiral separation to give biphenyl butyric
40 acid substituted on 4 position amino or derivative thereof. The product can be converted to sacubitril. The process of the method is depicted as follows:

